

Precautionary Savings and Stabilization Policy in a Present-Biased Economy

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This paper

Precautionary saving implications of Heterogeneous Agents (HA) models

- Idiosyncratic income risk + Incomplete markets → **precautionary saving**
 - ▶ Inconsistent with empirical result
 - ▶ The response of asset accumulation to variation in unemployment insurance (UI) schedules
- Introduction of **present bias** (along with naïvety)
 - ▶ procrastination in saving in the present
- Implications of present bias on **stabilization policy**
 - ▶ UI policy is less effective under a present-biased economy

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- 1 Model
- 2 Empirical Identification
- 3 Implications for Stabilization Policy (UI)
 - Exercise 1: Discretionary policy
 - Exercise 2: Automatic stabilizer
- 4 Conclusion
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Model - Bewley economy

- Discrete time (monthly)
- A unit mass of households i subject to **uninsurable income risk** and **unemployment risk**.
 - ▶ At the beginning of the period t

Employed $\rightarrow$ Unemployed (with prob s_t)
Unemployed $\rightarrow$ Employed (with prob f_t)
- Heterogeneity of households
 - ① Preferences: degrees of present bias
 - ② Employment status, labor productivity, and asset holdings

Households - Present Bias

- Preferences ▶ Quasi-Hyperbolic discounting

$$\mathcal{V}_{it} = u(c_{it}) - \nu(n_{it}) + \beta_i \mathbb{E}_t \sum_{k=1}^{\infty} \delta^k (u(c_{i,t+k}) - \nu(n_{i,t+k})) \quad (1)$$

$\begin{cases} \beta_i = 1 : \text{exponential/standard preference} \\ \beta_i < 1 : \text{present bias, time-inconsistent} \end{cases}$

where $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$, $\nu(n) = \psi \frac{n^{1+\varphi}}{1+\varphi}$ and $\beta_i \in \{\beta^L, \beta^H\}$

- Assume agents are *naïve*, falsely believing their future selves will not deviate from currently established plans. (O'Donoghue and Rabin, 1999) ▶ Bellman Equation

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- Assume agents are *naïve*, falsely believing their future selves will not deviate from currently established plans. (O'Donoghue and Rabin, 1999) ▶ Bellman Equation

Households - Unemployment risk

- Budget constraint

$$\begin{aligned} a_{it} + c_{it} &= (1 + r_t + \chi \cdot 1(a_{i,t-1} < 0))a_{i,t-1} + y_{it} \\ a_{it} &\geq \underline{a} \end{aligned} \tag{2}$$

- Earnings across employment status

$$y_{it} = \begin{cases} y_{it}^{labor} & \text{if employed} \\ y_{it}^{priv} + \mathcal{UI}_{it} & \text{if unemployed + receiving UI } (\zeta) \\ y_{it}^{priv} + T^{sn} & \text{if unemployed + not receiving UI } (1 - \zeta) \end{cases}$$

Households - Income risk

- Idiosyncratic labor productivity

$$z_{it} = \begin{cases} \mu^z + \rho^z z_{i,t-1} + \sigma^z \varepsilon_{it} & \text{if employed} \\ z_{i,t-1} & \text{if unemployed} \end{cases} \longrightarrow$$

- Earnings

$$\begin{aligned} y_{it}^{labor} &= (1 - \tau_t) w_t n_{it} e^{z_{it}} \\ \mathcal{UI}_{it} &= (1 - \tau_t) \min\{\textcolor{red}{b} w n e^{z_{it}}, \bar{b}\} \end{aligned}$$

* Replacement Rates **b**: expire after N^{UI} periods [► Distribution of replacement rates](#)

Firms/Government

- Perfectly competitive representative firm

$$Y_t^{firms} = Z \cdot N_t \quad (3)$$

- Government

$$B_t = (1 + r_t)B_{t-1} + G + S_t - R_t \quad (4)$$

$$R_t = \tau_t \left(\int y_{it}^{labor} 1_{it}^e di + \int \mathcal{UI}_{it} 1_{it}^{ui} di \right)$$

$$S_t = \int \mathcal{UI}_{it} 1_{it}^{ui} di + T^{sn} \int 1_{it}^{sn} di$$

R_t : total government revenue S_t : social security spending

Market Clearing

- Goods market:

$$C_t + G - \chi \int a_{i,t-1} 1_{it}^{debt} di = Y_t (= Y_t^{firms} + Y_t^{priv}) \quad (5)$$

- Labor market:

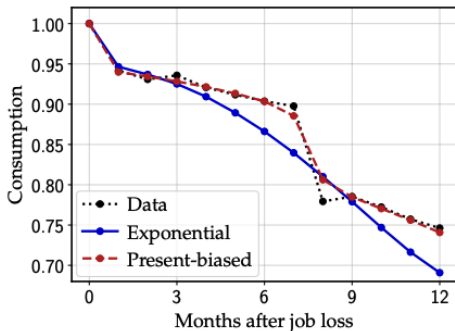
$$N_t = E_t \cdot n_t \quad (6)$$

- Asset market: government is the sole issuer of bonds in the economy

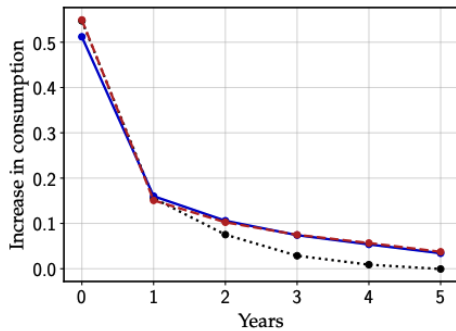
$$\int a_{it} di = B_t \quad (7)$$

Calibration

- Internal calibration to match 2 sets of empirical moments
 - ① Spending during unemployment of UI recipients (Ganong and Noel, 2019)
 - ② Intertemporal MPC (iMPCs) (Auclert et al., 2018)



(a) Consumption during unemployment



(b) iMPCs

Calibration -ctd

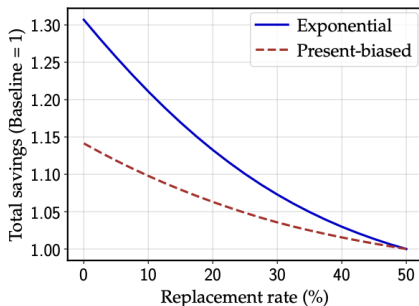
Table 1: Calibrated parameters

Parameter	Meaning	Value		Source/Target
Externally calibrated parameters				
σ	CRRA	2		Carroll (1997)
ρ_z	AR(1) persistence	0.993		Floden and Linde (2001)
$\sigma_z/\sqrt{1-(\rho_z)^2}$	AR(1) std.	0.51		
s	Separation rate	0.012		U = 5.97%
f	Job finding rate	0.19		Avg. weeks unemp. 17
b	Replacement rate	0.50		US UI system
$\bar{b}$	UI cap	0.48		
N^{ui}	Duration	7		
ξ	Take-up	0.77		Auray et al. (2019)
r	Interest rate	4% p.a.		Standard
B/Y	Liq. assets	0.26		Kaplan, Moll and Violante (2018)
Internally calibrated parameters				
		Exponential	Present-biased	
δ	Discounting	0.988	0.991	Target interest rate
β_L, β_H	Present bias	1	0.5, 0.9	
ω_H	Share of high β	-	0.74	
y^{priv}	Non-UI income	0.320	0.320	
T^{sn}	Safety net transfers	0.087	0.087	iMPC and consumption during unemployment
$\underline{a}$	Borrowing constraint	0	-1.48	Government budget balance
χ	Borrowing cost	-	0.0098	
τ^l	Labor income tax	0.217	0.217	

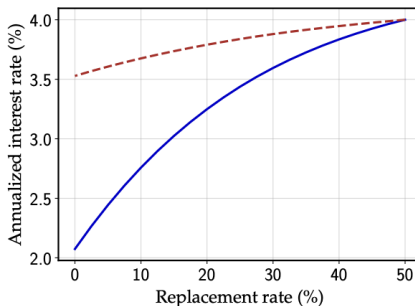
Result

$$\mathcal{UI}^* = (1 - \tau) \min\{bwne^{z_i^*}, \bar{b}\}$$

The experiment is to scale down both the replacement rate b and the maximum benefit $\bar{b}$ by the same factor until reaching zero



(a) Partial equilibrium



(b) General equilibrium

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Empirical Identification

- **Specification** (Two-Way Fixed Effects OLS, Engen and Gruber 2001)

$$\left(\frac{\text{Liquid Assets}}{\text{Earnings}} \right)_{i,s,t} = \alpha_s + \tau_t + \gamma \times RR_{s,t} + X'_{i,s,t} \beta + \varepsilon_{i,s,t} \quad (8)$$

- γ : measures the change in the levels of liquid assets to income ratios following a 10 p.p. change in replacement rates $RR_{s,t}$

- **Data**

Survey of Income and Program Participation (SIPP)

- 1984-2008 panels: 50 states, 1984-2011
- household balance sheets, demographics, earnings, and state of residence, social insurance payments
- **Sample restriction: HHs with stable labor earnings histories (aged 25-64)**

Empirical Identification - ctd

$$\left(\frac{\text{Liquid Assets}}{\text{Earnings}} \right)_{i,s,t} = \alpha_s + \tau_t + \gamma \times \textcolor{red}{RR}_{s,t} + X'_{i,s,t} \beta + \varepsilon_{i,s,t}$$

- **Simulated policy instruments** (Currie and Gruber, 1996)

Construct measures of UI generosity $RR_{s,t}$ by

- ▶ Strip sample of household heads from state and time information: national cross-time sample
- ▶ Assume all such heads lose their jobs in a particular state-time cell and compute their eligibility/benefits using the UI schedule from that particular state-time cell
- ▶ Calculate median eligibility and replacement rates for the particular cell
- ▶ Repeat for all states and time periods (1984 H1:2011H2)

Result [▶ Comments](#)

Table 2: Two-way fixed effects regressions

	Net Financial Assets		
	(1)	(2)	(3)
Median RR	-0.0051 (0.0299)	-0.0391 (0.0261)	0.0109 (0.0298)
Outcome mean	0.06	0.06	0.06
N	178119	178119	178119
Individual controls	✓	✓	✓
State controls		✓	✓
Time FE	✓		✓
State FE	✓		✓

- Column 3: two-standard error lower band rules out decreases in asset-income ratios smaller than -0.005 per 10 p.p (\$350 at the sample mean for household earnings)
- Robustness check: (1) measure of liquid assets (2) IV & Stacked Regressions [▶ Robustness check](#)

Bringing together model and data

- Run the same reduced-form regressions in model-generated data using both the exponential and present-biased calibrations
- Allow time-varying replacement rates in the model: b_t follow a finite-state Markov chain

$$b \in \{0.25, 0.3, 0.35, 0.4, 0.45, 0.5, 0.55\}$$

- ▶ with prob p^{ref} , UI reform $\rightarrow b_t$ jumps to a neighboring point (increasing or decreasing with equal probabilities)
 - ▶ UI cap $\bar{b}_t$ time varying accordingly
- Draw repeated cross-sections of households every year (51 states for 22 years with sample 180 households)

Bringing together model and data

Table 3: Regressions in model-simulated data

	Data			Exponential	Present-biased
	NFA	GFA	Excl. risky		
Median RR	0.0109 (0.0298)	0.0229 (0.0238)	-0.0066 (0.0233)	-0.0636 (0.0267)	-0.0207 (0.0525)
Lower 95% confidence band	-0.0475	-0.0237	-0.0523	–	–

- Exponential model delivers substantially larger coefficients than the empirical estimates, outside confidence intervals [▶ Robustness check](#)
- Present-biased model delivers much lower coefficients that lie within confidence intervals

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Introducing Business Cycles - HANK

- Wage stickiness (Auclert et al. 2018)
 - ① A continuum of labor unions j set nominal wages (no present-bias)
 - ② Calvo rigidity (θ can partially index their wages to past inflation)

$$W_{jt} = W_{j,t-1} \left(\frac{P_{t-1}}{P_{t-2}} \right)^\iota$$

- ③ NKPC with indexation [▶ Derivation](#)

$$\pi_t = \frac{\delta}{1+\iota} \mathbb{E}_t \pi_{t+1} + \frac{\iota}{1+\delta\iota} \pi_{t-1} + \frac{(1-\theta)(1-\delta\theta)}{\theta(1+\delta\iota)} d \log \mu_t + \varepsilon_t^w$$

$$\mu_t = \frac{\mathcal{V}_t}{\mathcal{U}_t} = \frac{v'(n_t)}{\frac{1}{E_t} \int [u'(c_{it})(1-\tau_t)z_{it}w_t] 1_{it}^e di} : \text{Aggregate Labor Wedge}$$

Introducing Business Cycles - HANK

- Labor demand: $N_t = E_t n_t$.

$$d \log E_t = \lambda \cdot d \log N_t$$

$$d \log n_t = (1 - \lambda) \cdot d \log N_t$$

$$ds_t = -\eta \cdot df_t$$

- Monetary and fiscal policies

- ▶ Taylor rule

$$i_t = \phi^i i_{t-1} + (1 - \phi^i)(i + \phi^\pi \pi_t) + \varepsilon_t^i$$

- ▶ Real interest rate

$$1 + r_t = \frac{1 + i_{t-1}}{1 + \pi_t}$$

- ▶ Labor tax rule:

$$dR_t = \phi^B \cdot dB_{t-1}$$

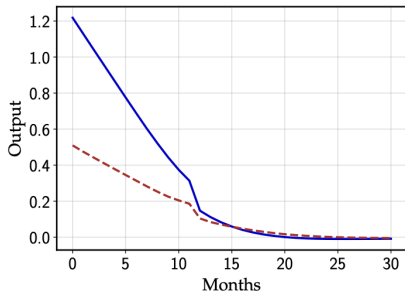
Estimation of HANK

- ① Calibration for labor demand ▶ Parameters for labor demand variation
 - ▶ λ : Use aggregate data on total hours, avg weekly hours, employment in the U.S nonfarm business sector (1960-2019 sample), a value of 0.817
 - ▶ η : use Shimer (2012) data (1960-2008), a value of 0.0142
 - ▶ Policy rule coefficients: $\phi^\pi = 1.5$, $\phi^B = 0.1$ (Auclert et al., 2020)
- ② Estimate nine parameters using Bayesian inference ▶ Parameters for Bayesian
 - ▶ Wage markup shock ε_t^w (ρ^w, σ^w)
 - ▶ Monetary shock ε_t^i (ρ^i, σ^i)
 - ▶ Discount factor shock ε_t^d (ρ^d, σ^d): $\delta \cdot \exp(\varepsilon_t^d)$
- ③ Use three time series (i) aggregate consumption growth (ii) inflation (iii) the Fed funds rate for an empirically realistic mix of shocks
- ④ Solve the linearized model using the method by Auclert et al. (2021) - Sequence-space jacobian

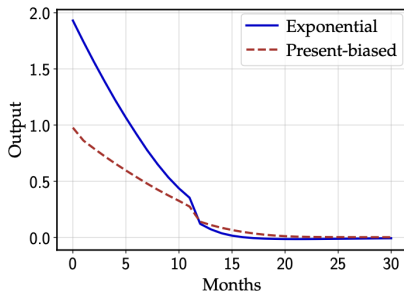
Simulation 1 - Temporary UI extensions

Two simulations: Start from the steady state. Announce transfer T to unemployed agents for 12 months (from $t=0$ to $t=11$)

- 1 *All unemployed* agents receive the transfer
- 2 Only unemployed agents with *expired benefits* receive the transfer



(a) Transfer to all unemployed

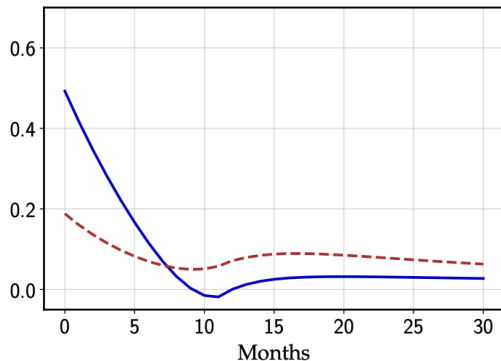


(b) Duration increase

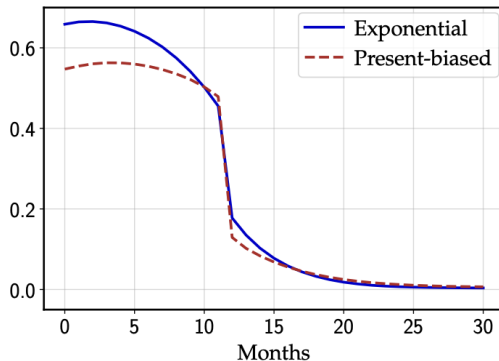
Simulation 1 - Temporary UI extensions

[► Comments](#)

Figure 4: Consumption responses to transfers to all unemployed agents (PE)



(a) Employed agents



(b) Unemployed agents

Simulation 2 - UI as an automatic stabilizer

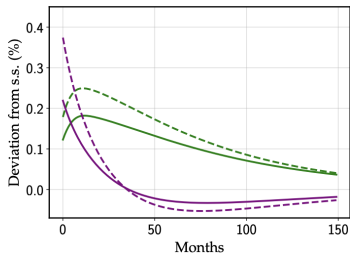
- Compute a new steady state with $b, \bar{b}$ reduced by half (same for T^{sn})
- One of the estimated three shocks ($\varepsilon_t^i, \varepsilon_t^w, \varepsilon_t^d$) or combined hit the economy around the new steady state
- Compute the increase in aggregate consumption volatility

Table 4: The effects of reduced automatic stabilizers on aggregate consumption volatility

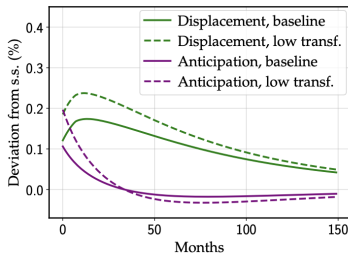
	Increase in the standard deviation of $\log(C_t)$				Annualized r
	All shocks	Only monetary	Only markup	Only discount	
Exponential	17.2%	9.0%	14.1%	19.6%	3.10%
Present-biased	10.1%	6.8%	10.9%	11.2%	3.72%

Simulation 2 - UI as an automatic stabilizer

- Agg. consumption response to Δ in employment ($d \log E_t = 0.978^t$)
- Decomposition of consumption response (first-order approximation)
 - ① *displacement effect*: changes in s_t, f_t but keeping fixed at s.s. households' expectations of these probabilities
 - ② *anticipation effect*: expected changes in f_t, s_t but keeping their actual values fixed at their s.s. levels



(a) Exponential



(b) Present-biased

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Conclusion

- In HA, PB **weakens** precautionary saving motives $\therefore$ procrastination in the present for self-insurance
- Empirical finding: a small & statistically non-significant effect of the changes in income risk on the sensitivity of precautionary savings
 $\Rightarrow$ **Rejects** the exponential model vs. **In favor of** the PB model
- UI as a stabilization policy **less effective** under PB due to attenuation of precautionary saving $\Rightarrow$ Open policy questions

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Comments

- **Validity of the empirical finding**

- ▶ Many states have very frequent reforms in UI schedules. Possible that hhs are not fully informed of the policy
⇒ unanticipated vs. anticipated income shock ▶ Replacement Rates
- ▶ Do employed households actually care about UI benefits?: sample selection
- ▶ The model doesn't produce enough variation in income risk for the unemployed agents.

- **Inconsistent calibration to compare two models**

- ▶ Implicitly match the distribution of liquid asset by having lower EIS (β^L) vs. higher EIS (β^H)
- ▶ Avg ratio of consumption vs. Distribution of liquidity constrained HHs (Exponential vs. PB)

▶ Simulation 1

- **Why important to think about PB time preference?**

- ▶ Potency of monetary/fiscal policy
- ▶ Would be interesting to see the transmission mechanism of monetary policy with two-asset structures e.g 1) mortgage refinancing channel. e.g.2) inflation dynamics + redistribution channel

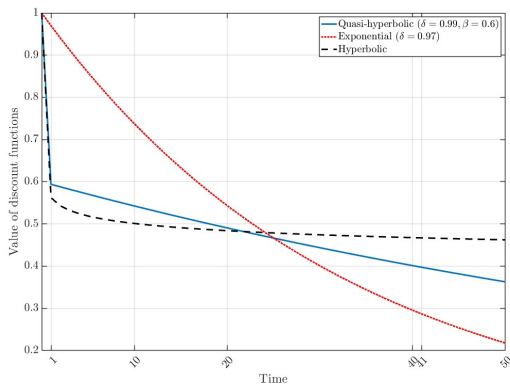
Outline

6 Supplementary

Quasi-Hyperbolic Discounting - Laibson (1999), Harris and Laibson (2001)

- Capture discount rates are much **greater in the short-run** than in the long-run.
- Discrete-time quasi-hyperbolic discount functions: $1, \beta\delta, \beta\delta^2, \dots, \beta\delta^t, \dots$ [◀ Go back](#)

Figure 5: Quasi-hyperbolic vs. Exponential Discount function



Bellman Equation for Naïve agents

◀ Household - present bias

- Exponential agents

$$V^{exp}(a) = \max u(c) + \delta \mathbb{E} V^{exp}(a')$$

$$\text{s.t. } c + a' = (1 + r)a + y, a' \geq 0$$

- Naïve present-biased agents

$$V(a) = \max u(c) + \beta \delta \mathbb{E} V^{exp}(a')$$

$$\text{s.t. } c + a' = (1 + r)a + y, a' \geq 0$$

Bellman Equation for Naïve agents

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$$V(a) = \max u(c) + \beta \delta \mathbb{E} V^{exp}(a')$$

$$\text{s.t. } c + a' = (1 + r)a + y, a' \geq 0$$

Zero-Liquidity Model - Bellman Equation ◀ Household - present bias

- Employed exponential agents

$$V_e^{exp}(a) = \max u(c) + \delta [(1-s)V_e^{exp}(a') + sV_u^{exp}(a')] \\ \text{s.t. } c + a' = (1+r)a + y_e, a' \geq 0$$

- Unemployed exponential agents

$$V_u^{exp}(a) = \max u(c) + \delta [fV_e^{exp}(a') + (1-f)V_u^{exp}(a')] \\ \text{s.t. } c + a' = (1+r)a + y_u, a' \geq 0$$

- Employed naïve present-biased agents

$$V_e(a) = \max u(c) + \beta\delta [(1-s)V_e^{exp}(a') + sV_u^{exp}(a')]$$

- Unemployed naïve present-biased agents

$$V_u(a) = \max u(c) + \beta\delta [fV_e^{exp}(a') + (1-f)V_u^{exp}(a')]$$

Zero-Liquidity Model - Euler Equation

Lemma 1

The unconstrained present-biased agent consumes according to the Euler equation

$$u'(c_e(a)) = (1 + r)\beta\delta [su'(c_e^{exp}(a')) + (1 - s)u'(c_u^{exp}(a'))] \quad (9)$$

The zero liquidity equilibrium is then characterized by the interest rate r such that the employed agents with zero assets choose to consume all their income ($c_e(0) = y_e$), but the Euler equation still holds with equality

Zero Liquidity parametrization: Werning (2015)

- Agents cannot borrow ($a = 0$)
- Supply of liquid assets shrinks to zero: $B \rightarrow 0$
- No heterogeneity in labor productivity ($e^{z_{it}}=1$), present bias $\beta^H = \beta^L = \beta$
- UI: sole source of income for the unemployed ($y^e = 1 - \tau_t, y^u = (1 - \tau_t)b$)

Proposition 1

$$\frac{dr}{db} = \underbrace{\beta}_{\text{discounting}} \delta(1+r)^2 \left[s\sigma b^{-\sigma-1} + \underbrace{(1-s)\sigma \left(\frac{c_e^{\text{exp}}}{y_e} \right)^{-\sigma-1} \frac{d}{db} \left(\frac{c_e^{\text{exp}}}{y_e} \right)}_{\text{time inconsistency}} \right]$$

△ Extra discounting

- Exponential preference:

$$\frac{dr}{db} = \delta(1+r)^2 s\sigma b^{-\sigma-1} = 0.046$$

△ Time inconsistency effect

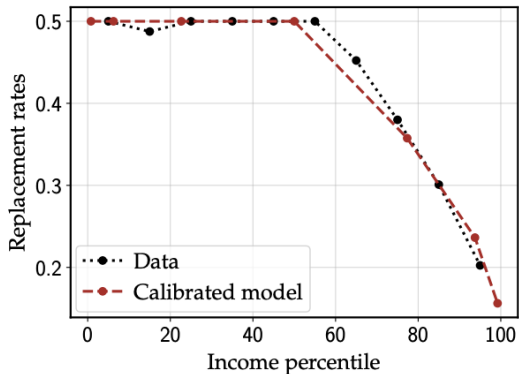
- Present-biased preference:

$$\frac{dr}{db} = 0.0037(0.046, -0.042)$$

Calibration - Replacement rates [◀ Go back](#)

The UI parameter values are set to match the distribution of replacement rates in the SIPP data.

Figure 6: Replacement rates along the income distribution



Robustness Check [◀ Go back](#)

	Baseline	GFA	Excl. risky	Elig. adj.	< Granular	No trim	Trim 10%	Time cluster
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Median RR	0.0109 (0.0298)	0.0229 (0.0238)	-0.0066 (0.0233)	0.0100 (0.0293)	0.0010 (0.0364)	-0.2432 (0.2031)	0.0132 (0.0241)	0.0109 (0.0294)
Outc. mean	0.06	0.10	0.01	0.06	0.06	0.17	0.04	0.06
N	178119	177812	176024	178119	178119	187504	168731	178119

Table 5: TWFE Robustness

Notes: this table reports estimates of regressions of net financial assets (normalized by annual household earnings) on simulated unemployment insurance median replacement rates. Each column on the table corresponds to a different robustness check. Column (1) corresponds to the results in Column (3) of Table 2, our baseline specification. Column (2) replaces NFA with Gross Financial Assets as the outcome. Column (3) excludes stocks and mutual funds from NFA. Column (4) adjusts replacement rates for eligibility. Column (5) instead uses a less granular, state-time level measure of UI generosity rather than a state-time-number of kids measure. Column (6) reports results without any trimming of the dependent variable. Column (7) repeats Column (6), but trimming the top and bottom 5% of the outcome distribution. Column (8) reports results with double-clustered standard errors at the state and year level. Except for column (8), standard errors are clustered at the state level.

Robustness Check [◀ Go back](#)

	Exponential model							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	$\sigma = 1$	Varying f	Low liq.	High liq.	Borrowing	$y^{priv} = 0$	$N^{ui} = \infty$	$\xi = 1$
Median RR	-0.0551	-0.0589	-0.0611	-0.0607	-0.0802	-0.1885	-0.0842	-0.0903
	(0.0315)	(0.0267)	(0.0122)	(0.0423)	(0.0295)	(0.0294)	(0.0285)	(0.0285)

Table 6: Regressions for variants of the exponential model

- **UI calculator from Kuka (2020)**

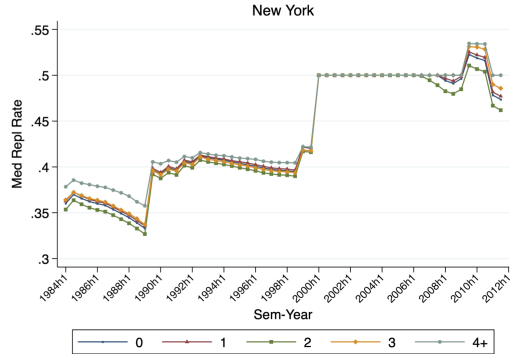
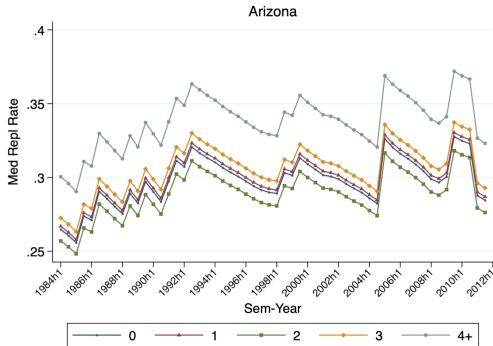
semesterly information on the particular formulae each state uses to compute benefits or eligibility

- ▶ Using information on earnings history and number of children, the calculator outputs how much each household head would make in benefits if she were to lose her job in her current state of residence
- ▶ Convert weekly benefits into a replacement rate by dividing by a (weekly) measure of previous earnings used in many states' UI formulae

Data appendix- Replacement rates

◀ Go back

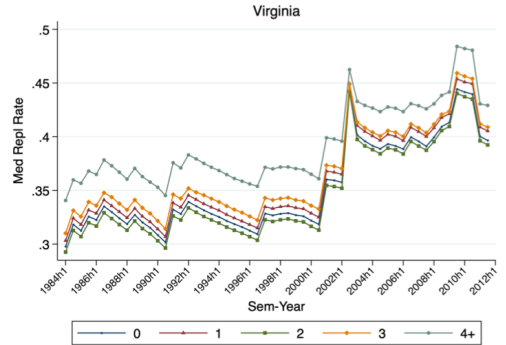
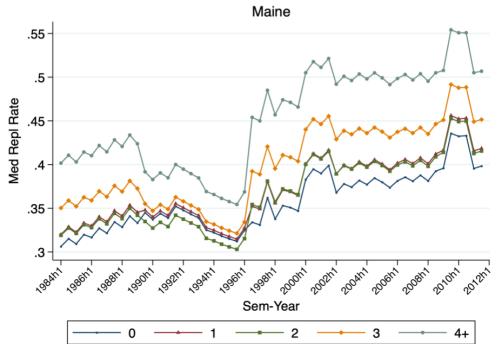
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Data appendix- Replacement rates

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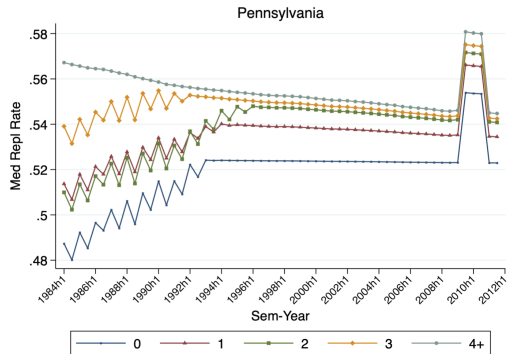
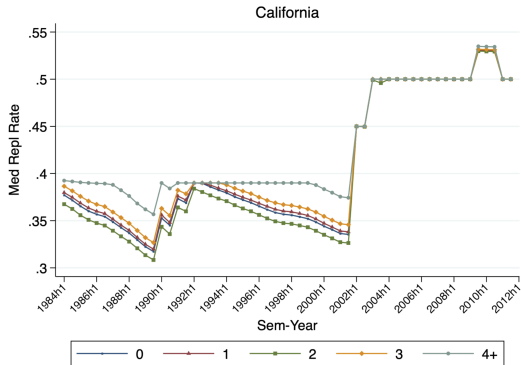
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Data appendix- Replacement rates

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◀ Comments



Data appendix- Summary Statistics

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Table 7: Summary Statistics

	Mean	SD	10th	25th	50th	75th	90th
<i>Panel A. Simulated instruments</i>							
Mean eligibility	0.97						
Median RR	0.42	0.08	0.32	0.37	0.42	0.49	0.52
<i>Panel B. Households balance sheets</i>							
NFA / Earnings	0.06	0.32	-0.19	-0.04	0.00	0.06	0.35
Net financial assets	6647	33490	-10423	-1942	0	3960	26877
Interest bearing	7795	19777	0	0	757	6073	20933
Stocks	3934	20876	0	0	0	0	2993
Home equity	61279	98456	0	0	22316	86140	178940
Total wealth	110934	183392	781	7172	44121	138977	298781
Unsecured debts	5596	16654	0	0	1018	6238	16199
Secured debts	68897	108170	0	0	23294	108071	191580
Net Worth	106578	183954	-1324	3912	39160	132849	2995978

Derivation of NKPC with indexation ◀ Business Cycle

- Total labor input $N_t = \left[\int N_{jt}^{1-1/\nu_t} \right]^{\frac{\nu_t}{\nu_t-1}}$ yields $N_{jt} = \left(\frac{W_{jt}}{W_t} \right)^{-\nu} N_t$
- $N_{jt} = E_t n_{jt}$ with a slight abuse of notation
- Unions maximize

$$\mathbb{E}_t \sum_{k=0}^{\infty} \delta^k \frac{1}{E_{t+k}} \int [u(c_{i,t+k}) - v(n_{t+k})] 1_{i,t+k}^e di$$

$$= \frac{\partial y_{i,t+k}}{\partial W_{jt}^*}$$

- FOC: $\mathbb{E}_t \sum_{k=0}^{\infty} (\delta\theta)^k \frac{1}{E_{t+k}} \int [u'(c_{i,t+k}) \underbrace{\frac{\partial c_{i,t+k}}{\partial W_{jt}^*}}_{\frac{\partial n_{i,t+k}}{\partial W_{jt}^*}} - v'(n_{t+k}) \frac{\partial n_{i,t+k}}{\partial W_{jt}^*}] 1_{i,t+k}^e di = 0$

Real income of employed workers in $t+k$:

$$\begin{aligned} y_{i,t+k} &= (1 - \tau_{t+k}) \frac{\Pi_{t-1,t+k-1}^\ell W_{j,t}^*}{P_{t+k}} n_{t+k} z_{i,t+k} \\ &= (1 - \tau_{t+k}) z_{i,t+k} \frac{N_{t+k}}{E_{t+k}} \frac{\Pi_{t-1,t+k-1}^\ell W_{j,t}^*}{P_{t+k}} \left(\frac{\Pi_{t-1,t+k-1}^\ell W_{jt}^*}{W_{t+k}} \right)^{-\nu} \end{aligned}$$

Derivation of NKPC with indexation - Ctnd ◀ Business Cycle

● Partial derivatives

$$\begin{aligned}
 \frac{\partial y_{i,t+k}}{\partial W_{jt}^*} &= (1-\nu)(1-\tau_{t+k})z_{i,t+k}n_{t+k} \frac{(\Pi_{t-1,t+k-1}^\ell)^{1-\nu}}{P_{t+k}} \left(\frac{1}{W_{t+k}}\right)^{-\nu} (W_{jt}^*)^{-\nu} \\
 &= (1-\nu)(1-\tau_{t+k})z_{i,t+k}n_{t+k} \frac{W_{t+k}}{P_{t+k}} \left(\frac{\Pi_{t-1,t+k-1}^\ell}{W_{t+k}}\right)^{1-\nu} (W_{jt}^*)^{-\nu} \\
 &= (1-\nu)(1-\tau_{t+k})z_{i,t+k}n_{t+k}w_{t+k} \left(\frac{\Pi_{t-1,t+k-1}^\ell}{W_{t+k}}\right)^{1-\nu} (W_{jt}^*)^{-\nu} \\
 \frac{\partial n_{t+k}}{\partial W_{jt}^*} &= -\nu n_{t+k} \left(\frac{\Pi_{t-1,t+k-1}^\ell}{W_{t+k}}\right)^{-\nu} (W_{jt}^*)^{-\nu-1}
 \end{aligned}$$

● FOC simplifies as:

$$\mathbb{E}_t \sum_{k=0}^{\infty} (\delta\theta)^k \left(\frac{\Pi_{t-1,t+k-1}^\ell}{W_{t+k}}\right)^{-\nu} \left\{ \frac{\Pi_{t-1,t+k-1}^\ell}{W_{t+k}} W_{jt}^* \mathcal{U}_{t+k} - \frac{\nu}{\nu-1} \mathcal{V}_{t+k} \right\} = 0$$

where $\mathcal{U}_t = \frac{1}{E_t} \int [u'(c_{it})(1-\tau_t)z_{it}w_t] 1_{it}^e di$, $\mathcal{V}_t = v'(n_t) \implies \mu_t = \frac{\mathcal{V}_t}{\mathcal{U}_t}$

Derivation of NKPC with indexation - Ctnd ◀ Business Cycle

- In a zero-inflation S.S: $\mathcal{U} = \frac{\nu}{\nu-1} \mathcal{V}$
- Log-linearize FOC around the zero-inflation ss to get:

$$\mathbb{E}_t \sum_{k=0}^{\infty} (\delta\theta)^k \left(1 + (1-\nu)\iota\pi_{t-1,t+k-1} - (1-\nu)\widehat{W}_{t+k} \right) \{ \widehat{W}_t^* + \iota\pi_{t-1,t+k-1} - \widehat{W}_{t+k} + \widehat{\mathcal{U}}_{t+k} - \widehat{\mathcal{V}}_{t+k} \} = 0$$

$$\begin{aligned} \widehat{W}_t^* &= (1-\delta\theta)\mathbb{E}_t \sum_{k=0}^{\infty} (\delta\theta)^k (\widehat{W}_{t+k} + \widehat{\mathcal{V}}_{t+k} - \widehat{\mathcal{U}}_{t+k} - \iota\pi_{t-1,t+k-1}) = 0 \\ &= (1-\delta\theta)(\widehat{W}_t + \widehat{\mathcal{V}}_t - \widehat{\mathcal{U}}_t + \delta\theta\mathbb{E}_t(\widehat{W}_{t+1}^* - \iota\pi_t)) \end{aligned}$$

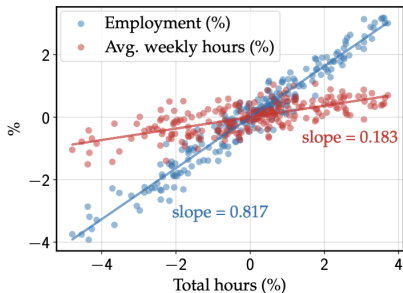
- LOM for agg. nominal wage

$$W_t^{1-\nu} = (1-\theta)(W_t^*)^{1-\nu} + \theta(W_{t-1}\Pi_{t-1}^\iota)^{1-\nu}$$

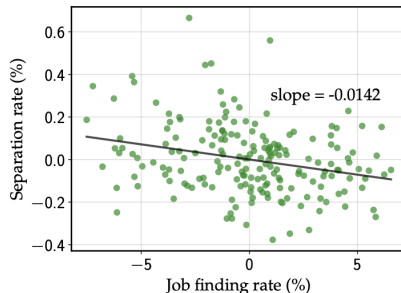
- $$\pi_t = (1-\theta)(\widehat{W}_t^* - \widehat{W}_{t-1}) + \theta\iota\pi_{t-1}$$

Calibrated parameters for labor demand variation ◀ Labor demand

- λ : Take logarithms of the quarterly time series data on employments/hours and extract their cyclical components using the HP filter with the standard smoothing parameter value of 1600
- η : Both job separation rate/finding rate are cyclical components from HP filter as above. The best fitting line is obtained from the first principal component.



(a) Hours worked and employment



(b) Separation and job finding rates

Priors and posteriors of estimated parameters ◀ Estimation

- The prior distributions are standard in the literature (Smets and Wouters, 2007)
- Draw 200,000 parameter values from the posterior distribution using the Metropolis-Hastings algorithm
- Estimation using the present-biased calibration only, use the same parameter values for both models

		Prior Distribution			Posterior Distribution		
		Distr.	Mean	St. Dev.	Mode	Mean	St. Dev.
ϕ^i	Taylor rule inertia	Beta	0.85	0.1	0.930	0.927	0.010
θ	Wage stickiness	Beta	0.85	0.1	0.982	0.982	0.003
ι	Wage indexation	Beta	0.5	0.2	0.589	0.591	0.064
ρ^d	Shock persistence	Beta	0.5	0.2	0.978	0.977	0.050
ρ^i		Beta	0.5	0.2	0.545	0.477	0.082
ρ^w		Beta	0.5	0.2	0.343	0.336	0.087
σ^d	Shock s.d.	Inv. Gamma	0.05	1	0.025	0.027	0.005
σ^i		Inv. Gamma	0.05	1	0.020	0.022	0.003
σ^w		Inv. Gamma	0.05	1	0.032	0.030	0.007

Figure 7: IRF to discount rate shock ε_t^d

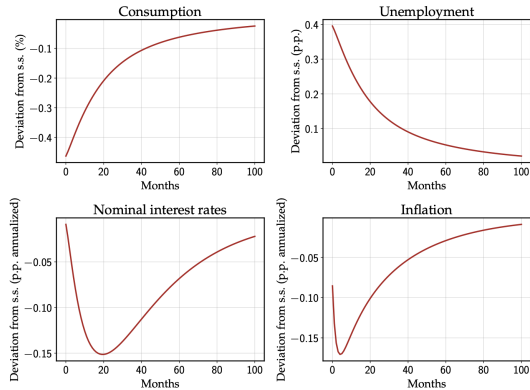


Figure 10: IRF to discount rate shock ε_t^d

Impulse response ◀ Estimation

Figure 8: IRF to monetary shock ε_t^i

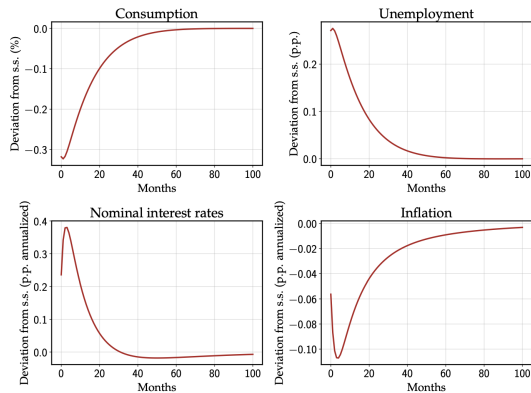


Figure 11: IRF to monetary shock ε_t^i

Figure 9: IRF to wage mark-up shock ε_t^w

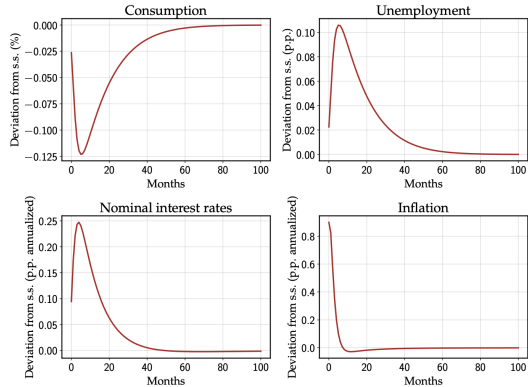


Figure 12: IRF to wage mark-up shock ε_t^w